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from nltk.tokenize import RegexpTokenizer
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer
import string
from sklearn.datasets import fetch_20newsgroups

# Download nltk data
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('punkt_tab')
nltk.download('wordnet_tab')

data = fetch_20newsgroups(subset='train', categories=('science', 'sports'), shuffle=True)
documents = data.data
print('Total documents:', len(documents))

stop_words = nltk.corpus.stopwords.get('english')
tokenizer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

def preprocess_text(text):
    text = text.lower()

    tokens = word_tokenize(text)

    tokens = [t for t in tokens if t not in stop_words]

    tokens = [t for t in tokens if t not in stop_words]
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tokens_list = [tokenizer.tokenize(t) for t in tokens_stream]

cleaned_text = '\n'.join(tokens_list)
return cleaned_text

sample = documents[0]
print('Original text (first 100 chars):', sample[:100])

cleaned = preprocess_text(sample)
print('Cleaned text (first 100 chars):', cleaned[:100])

# Process all documents
processed_docs = [preprocess_text(doc) for doc in documents]
print('Preprocessing completed for all dataset.')
```

[nltk data] Downloading package punkt to /root/nltk_data...

[nltk data] Package punkt is already up-to-date

[nltk data] Downloading package punkt_tab to /root/nltk_data...

[nltk data] Unzipping tokenizers/punkt_tab.zip

[nltk data] Downloading package stopwords to /root/nltk_data...

[nltk data] Package stopwords is already up-to-date

[nltk data] Downloading package wordnet to /root/nltk_data...

[nltk data] Package wordnet is already up-to-date

Total documents: 11114

Original text (first 100 chars):

I was wondering if anyone out there could enlighten me on this car I saw the other day. It was a 3-door sports car, looked to be from the late 60s/early 70s. It was called a Briskin. The doors were really small. In addition, the front bumper was separate from the rest of the body. This is all I know.

Cleaned text (first 100 chars):

wonder anyone could enlighten car saw day 3-door sport car look late 60s/early 70 call briskin door really small still front bumper separ rest body know anyone tells model name engine spec year product car make history white.

Preprocessing completed for full dataset.